Trace Boggs

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Education

- **Cameron University** — Lawton, OK
  Bachelors of Science in Computer Science; 4.0 GPA with Presidents Honor Roll — *August 2023 – May. 2026*
  & Bachelors of Arts in Mathematics; 4.0 GPA with Presidents Honor Roll.

Work Experience

- **Brunswick Navico Group** — Tulsa, Oklahoma
  Software Engineer Intern — *May 2025 - August 2025*
  - **Python GUI Development**: Built a desktop application enabling engineers to communicate with embedded devices over BLE.
  - **Automated Integration Tests**: Implemented tests to validate device configuration, verify firmware responses, and catch incorrect setting updates.
  - **Object-Oriented Architecture**: Designed modular classes for BLE communication, packet construction, and test execution to ensure maintainability.
  - **UX & Logging**: Added progress bars, real-time updates, and structured log files to support QA and debugging.
  - **Outcome**: Brought down regression testing time during development by over 95%, significantly increasing engineer productivity.
- **Cameron University** — Lawton, Oklahoma
  Mathematics Tutor — *January 2024 - Present*
  - **Tutored** students in Math and Computer Science subjects, including College Algebra, Trigonometry, Calculus, Programming, and Discrete Math.
  - **Led** both **one-on-one and group tutoring sessions**, helping students with coursework, problem-solving strategies, and exam preparation, resulting in improved homework and exam performance.
  - **Communicated** regularly with students to identify gaps in understanding, build strong study habits, and reinforce key concepts.

Research Experience

- **Math Capstone Research on Newton-Like Methods** — Cameron University
  Undergraduate Researcher — *August 2025-Present*
  - **Paper**: Assisted in writing a research paper titled *Developments on the Local Convergence of Newton-Like Methods in Normed Linear Spaces with Applications to Motion Problems.* Used LaTeX to write up the paper, designed applied examples of the theory, and helped in constructing the proofs of the theory.
  - **OK Research Day Presentation(Feb 2026)**: Designed a research poster to be presented in February of 2026 at Oklahoma Research Day.
  - **NCUR Presentation(April 2026)**: Designed a research poster to be presented in April 2026 at NCUR.
- **Neural Networks Independent Study** — Cameron University
  Undergraduate Researcher — *January 2025 – December 2025*
  - **Research Poster Presentation**: Conducted a literature review on the evolution of artificial intelligence and designed an educational research poster titled *Artificial Intelligence: Past, Present, and Future*, presented at Oklahoma Research Day 2025.
  - **Live-Inference Emotion Recognition Model**: Developed a CNN-based human emotion recognition system using transfer learning with MobileNetV2, fine-tuned on the FER-2013 dataset. Investigated performance degradation and prediction instability when transitioning from static image evaluation to real-time webcam inference.

Projects

- **Euler's Method Calculator** Sep 2024
  C++, Crow, HTML/CSS, JavaScript, SSL, Nginx
  - **Description**: Developed a full-stack web application to numerically solve ODEs with Euler's Method, hosted on a Raspberry Pi and accessible at `eulerscalculator.net`.
  - **Backend**: Built a C++ backend using Crow, implementing a RESTful API for all computations and optimizing frontend-backend communication.
  - **Error Handling**: Implemented robust validation and domain-checking for mathematical expressions.
  - **Deployment**: Configured SSL and reverse proxying with Nginx for secure public hosting.
- ***LightStop* Mobile Game** May 2024
  C#, Unity, APIs
  - **Description**: Developed and released a mobile game inspired by Cyclone/Pop-the-Lock, published on both iOS and Android.
  - **Features**: Integrated Unity Ads, implemented a live leaderboard, and designed gameplay mechanics and visual effects.
  - **Deployment**: Managed platform builds, store listings, and release pipeline for both platforms.
  - **Website**: sites.google.com/view/lightstop/home.
- **University PLUS Scholarship Website** Ongoing
  Leadership, Full-Stack Web-Dev
  - **Description**: Leading a team of university ACM members to design and develop a web application for the university. Coordinating development tasks, implementing features, and ensuring project progress toward completion next semester.

Achievements

- **Regional University Baccalaureate Scholarship Recipient**: Received the institution's top academic scholarship, which included a full tuition waiver and a significant stipend based on academic merit.

Technical Skills

- **Proficient**: C++, C, Unity/C#, Git, HTML/CSS/JS/PHP, LaTeX, AGILE Workflows
- **Comfortable**: Java, Python, Arduino, TensorFlow, Flutter, SQL, MATLAB